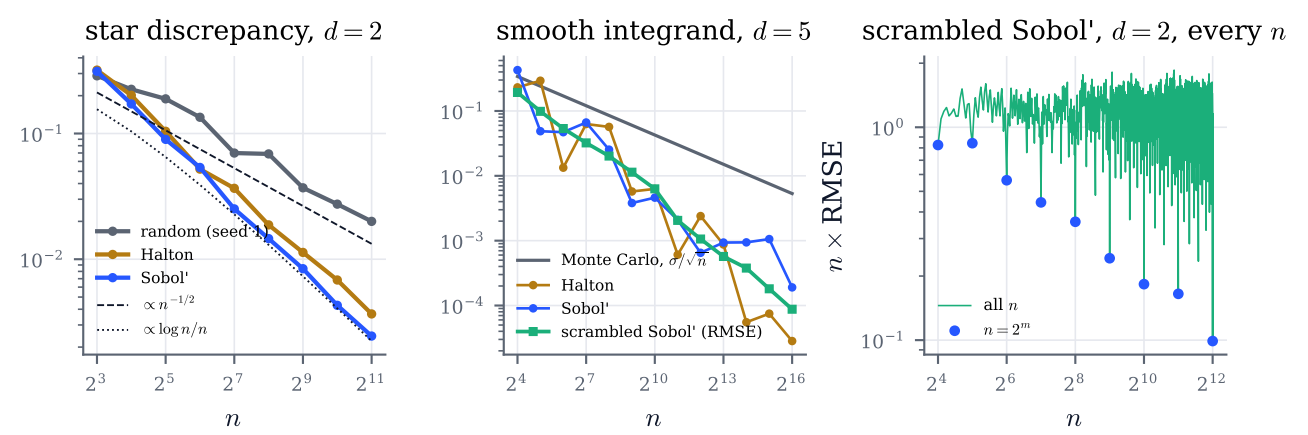
star discrepancy, $d = 2$
10^{-1}
10^{-2}
random (seed 1)
Halton
Sobol'
$\propto n^{-1/2}$
$\propto \log n / n$
2^3 2^5 2^7 2^9 2^{11}
n
smooth integrand, $d = 5$
10^{-1}
10^{-2}
10^{-3}
10^{-4}
Monte Carlo, $\sigma/\sqrt{n}$
Halton
Sobol'
scrambled Sobol' (RMSE)
2^4 2^7 2^{10} 2^{13} 2^{16}
n
scrambled Sobol', $d = 2$, every n
$n \times$ RMSE
10^0
10^{-1}
all n
$n = 2^m$
2^4 2^6 2^8 2^{10} 2^{12}
n